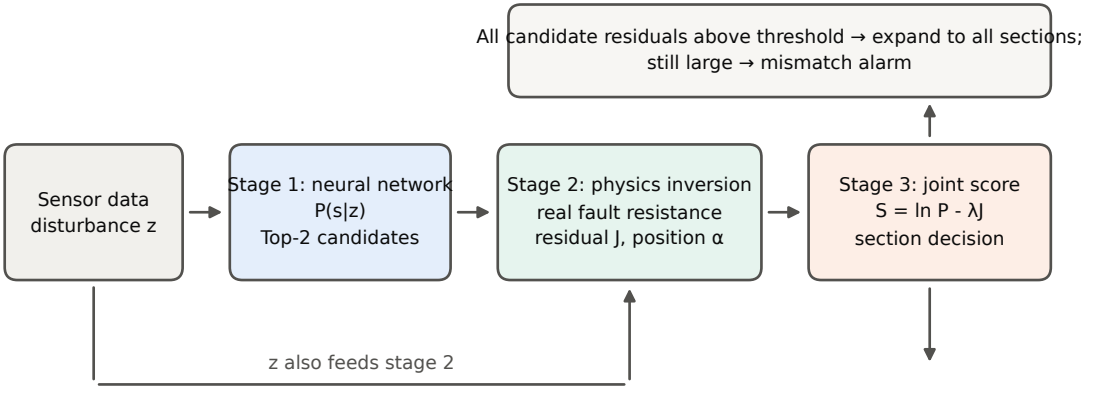
All candidate residuals above threshold → expand to all sections;
still large → mismatch alarm
Sensor data
disturbance z
Stage 1: neural network
P(s|z)
Top-2 candidates
Stage 2: physics inversion
real fault resistance
residual J, position α
Stage 3: joint score
S = ln P - λJ
section decision
z also feeds stage 2
Output: section, position, confidence